Sustainability Now!
Sustainability Concepts and Frameworks
Lecture 01

Bruce D. Marron

October 15, 2026

What is Sustainability?

Sustainability

Sustainability: Concepts and frameworks of stewardship, which are translated to direct human behaviors (decisions and management actions) and applied across multiple, interlinked complex adaptive systems (CAS), to ensure intergenerational equity.

The Goal: Flourishing of the Seven Generations.



Sustainability



Sustainability

Gro Harlem Brundtland, 41

First Woman Prime Minister of Norway, 1981



United Nations

Report of the World Commission on Environment and Development: Our Common Future

Transmitted to the General Assembly as an Annex to document A/42/427 - Development and International Co-operation: Environment

"...development that meets the needs of the present without compromising the ability of future generations to meet their own needs"

Sustainability

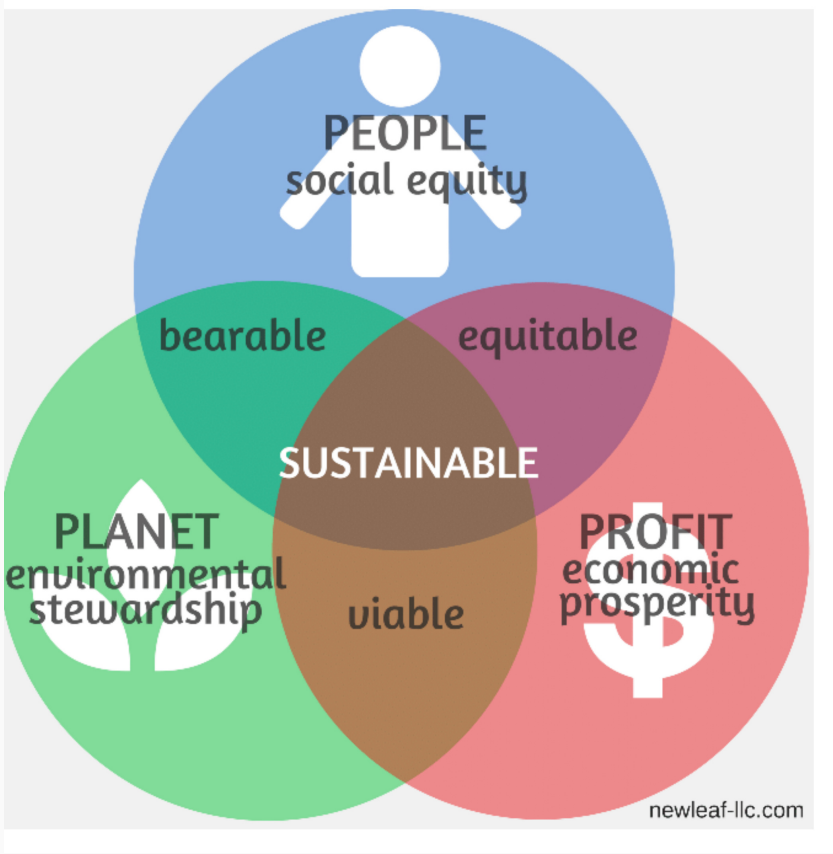
Brundtland says:

Wake up! (spiritual, social, environmental, economic, political awareness)

Cooperate! (empathetic, equity-based cooperation)

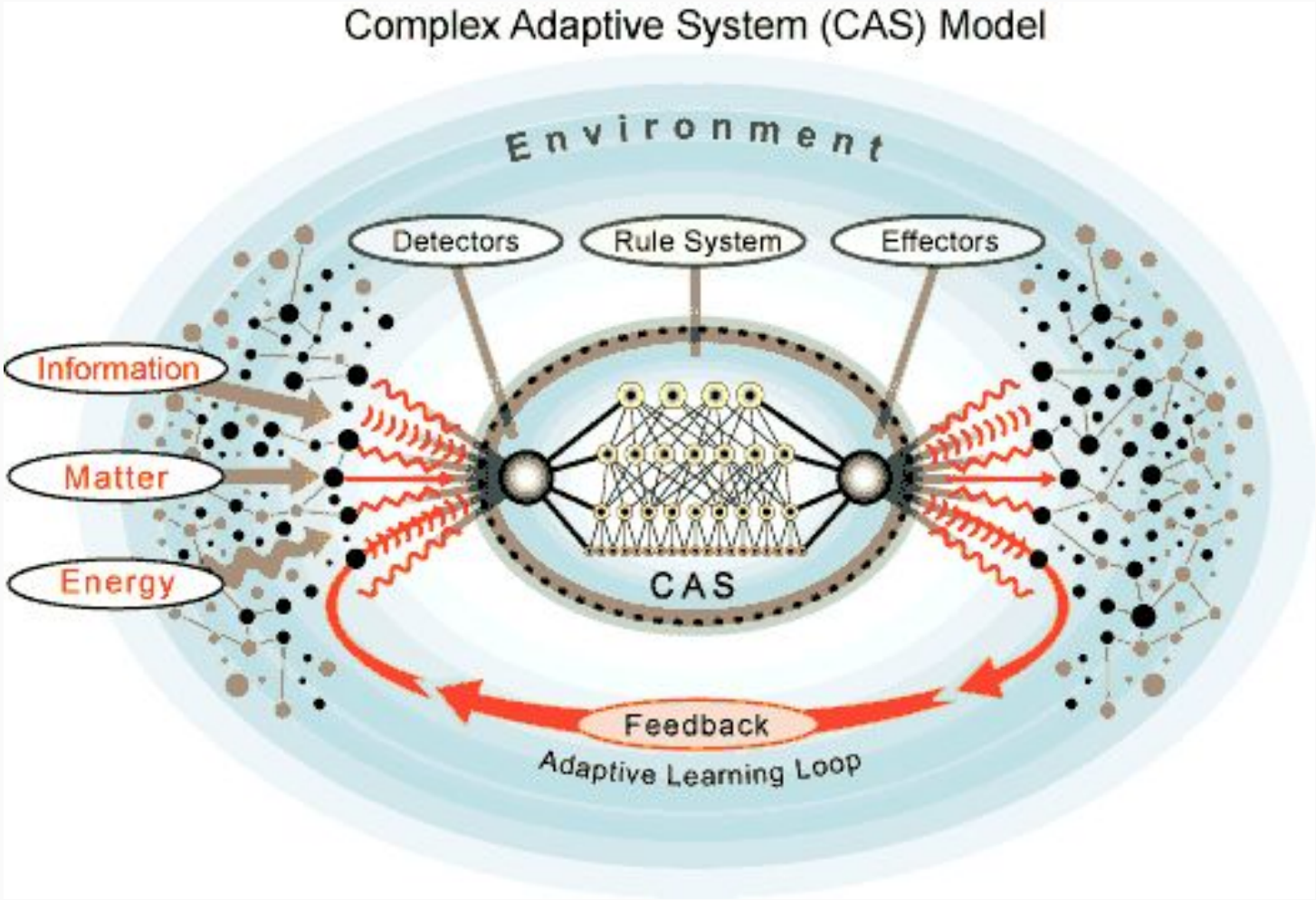
Break out! (imagine new systems and build them)

Sustainability



What are the glaring assumptions for this Triple Bottom Line?

Complex Adaptive Systems



Complex Adaptive Systems

Complex means...
Adaptive means ...
System means ...

...Emergence...
...Self-Organization...
...Feedback Loops...
...Resilience...

Complex Adaptive Systems



Complex Adaptive Systems



Resilience

Resilience is the use of dynamic human imagination to adapt, transform, evolve, innovate, self-organize, restructure, and survive when confronted with identity-level disruptive change from turbulence, emergence, collapse, devastation, or crisis

Reslience pathways may lead to adaptation (The Ship of Theseus) or transformation (Death of the Gold Standard)

Resilience pathways may lead to dead ends

"Sometimes you need to break the resilience of the current system to enable shifts away from the current pathway(s) into new ones, into alternative basins of attraction"